

JKBOSE 2026

CLASS 11TH

Morphology of Flowering Plants

LECTURE-09

BOTANY

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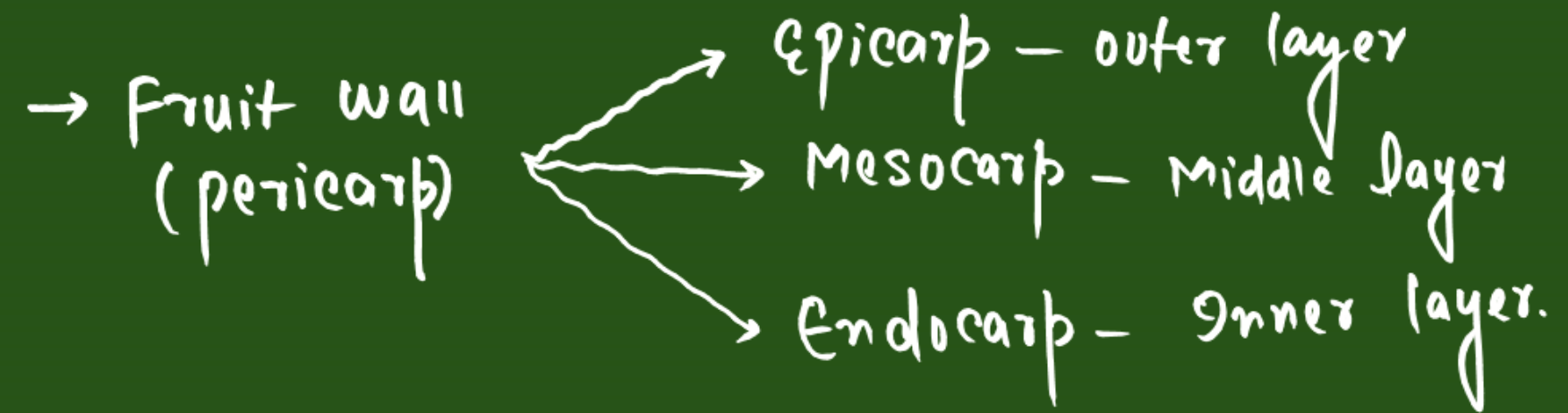


Topics To Covered

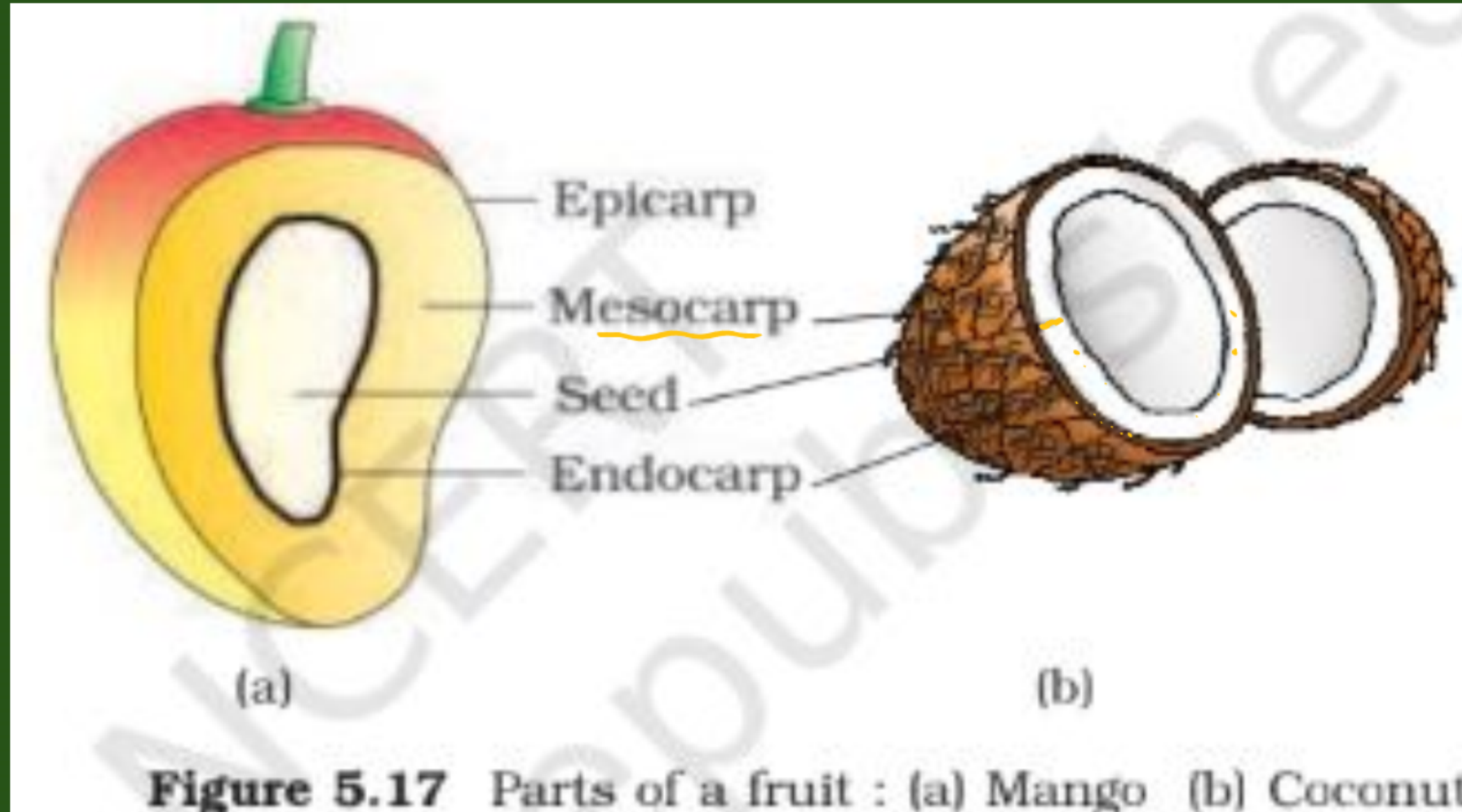
1 FRUIT (contd...)

2 Solonaceae Family

RECAP



FRUIT (Contd...)



FRUIT

03 TYPES

a. Eucarp / True Fruit

↗ EV = TRUE
 ↘ Carp = Fruit
 = Fruit which develops completely from ovary.
 = No other floral part is involved.
 = Ex. Mango.

b. Pseudocarp OR False Fruit

↗ pseudo = false
 ↘ Carp = Fruit
 = Fruit which develops from other floral parts like thalamus along with the development of ovary.
 Ex. Apple, Strawberry.

c. Parthenocarpic fruit

= These fruits are seedless.
 = These fruits are formed without fertilisation.
 Ex. = Banana, Vitis (grapes)

Note: Parthenocarpic fruits are produced with the help of Auxin.

Note

- ①. Mango and Coconut → develops from Monocarpellary Superior ovaries.
→ possess single seed.
ie, why these fruits are "Drupe"
- ② Coconut → Mesocarp is fibrous.



FAMILY SOLANACEAE

"Potato family"



Vegetative Characters:

Stem: herbaceous, rarely woody, aerial, underground stem in potato (Solanum tuberosum)

Leaves: alternate, simple, rarely pinnately compound, exstipulate; venation reticulate.

↓
Stipule -nt

Floral Characters:

Inflorescence: Solitary, axillary or cymose as in Solanum

→ **Flower:** bisexual, actinomorphic

→ **Calyx:** sepals five, united, persistent, valvate aestivation

⊕ ♂

$K_{(5)} \rightarrow$ lamiosepalous

→ **Corolla:** petals five, united; valvate aestivation

$C_{(5)} \rightarrow$ lamopetalous

→ **Androecium:** stamens five, epipetalous A_5

} $C_{(5)} A_5$

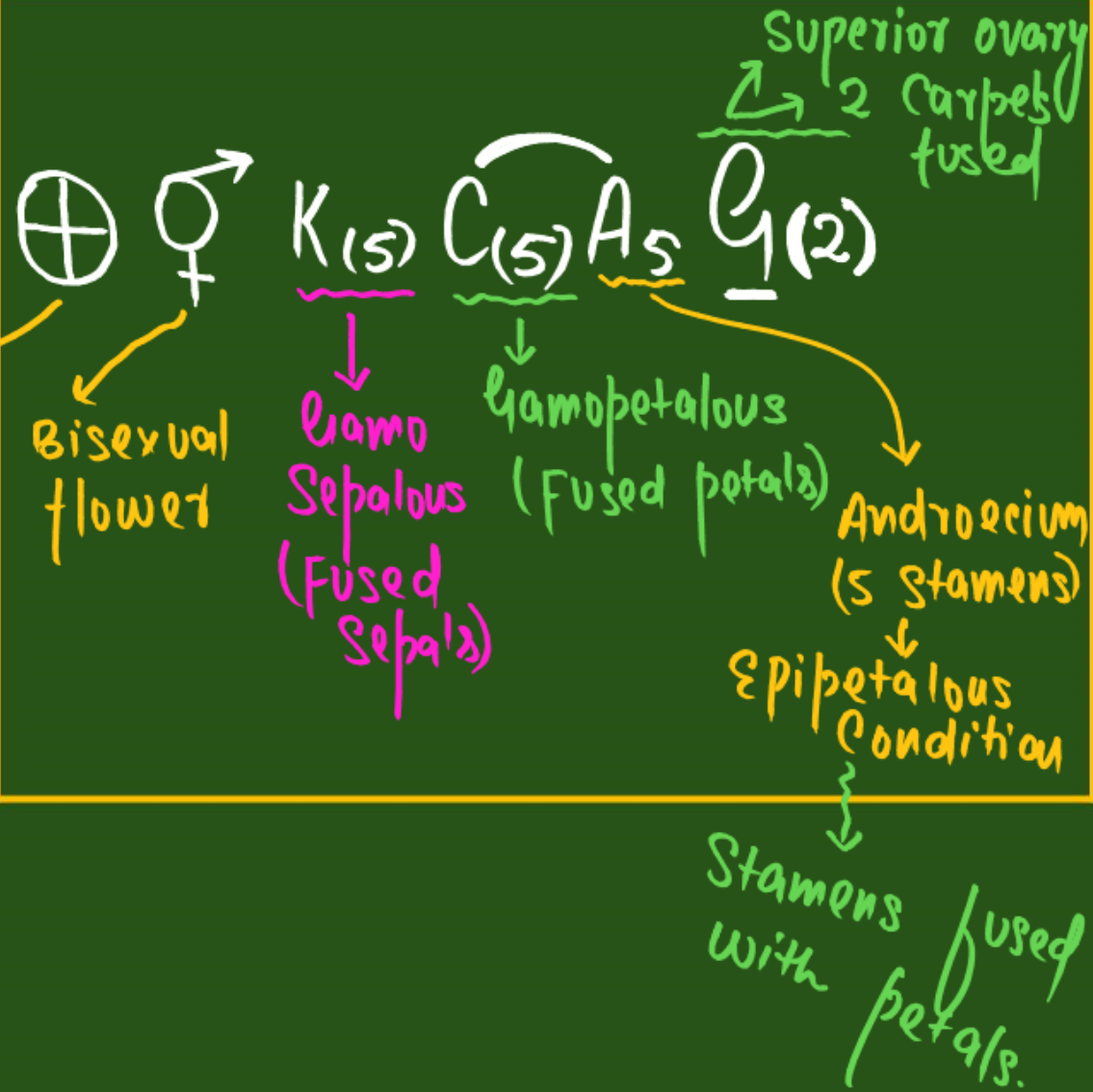
→ **Gynoecium:** bicarpellary (obliquely placed,) syncarpous; ovary superior, bilocular, axile placentation

$\underline{4}(2)$

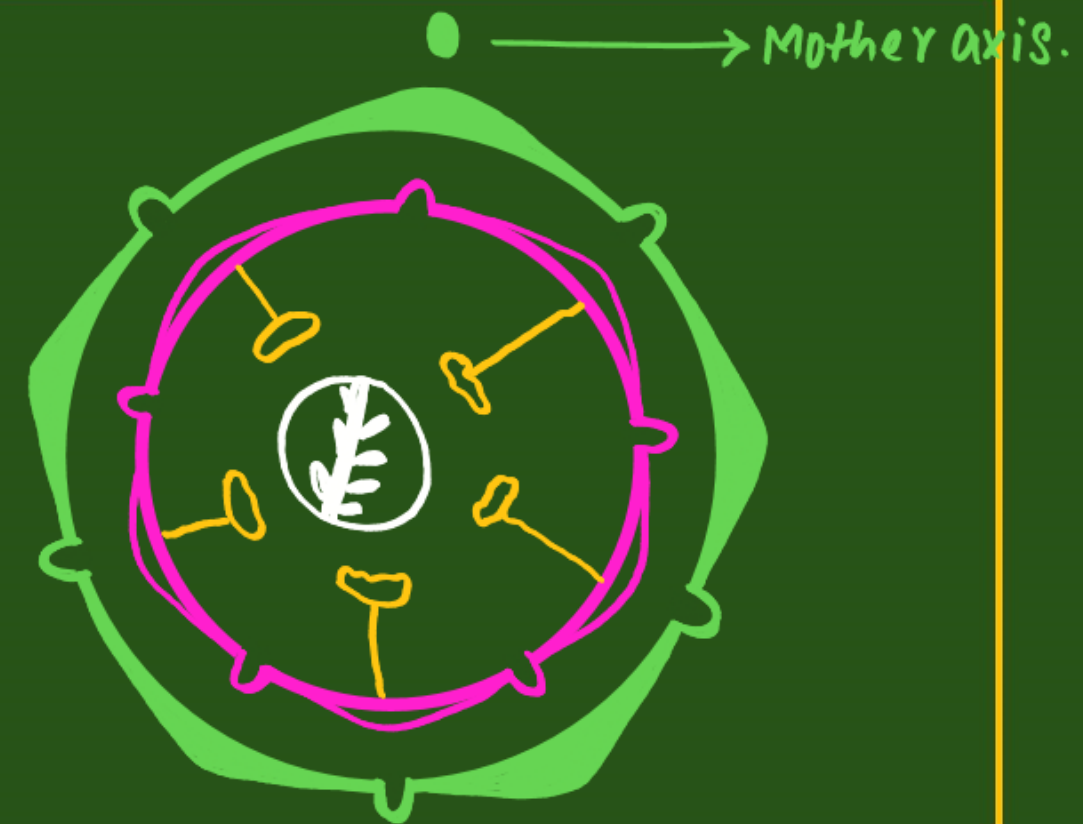
Fruits: berry or capsule

Solanacea Family

FLORAL FORMULA



FLORAL DIAGRAM



Economic Importance:

- ⇒ Source of food e.g. tomato, brinjal, potato
- ⇒ Spice e.g. chilli
- ⇒ Medicine e.g. belladonna, ashwagandha
- ⇒ Fumigatory e.g. tobacco
- ⇒ Ornaments e.g. petunia.

QUESTIONS

In some plants thalamus contributes to fruit formation. Such fruits are

[NEET-2020]

- A** Falsefruits
- B** Aggregate fruits
- C** True fruits
- D** Parthenocarpic fruit

QUESTIONS

In mango, the pericarp is well differentiated into

- A** Outer → epicarp
- B** Middle → mesocarp
- C** Inner → endocarp
- D** All of these

QUESTIONS

Select the correct statement w.r.t. mango and coconut:

- A** They develop from monocarpellary superior ovaries
- B** They develop from monocarpellary inferior ovaries
- C** They have fibrous epicarp
- D** They have fleshy edible mesocarp

Chapter complete...

Thank
You